

Target nomination from single-variant cis-eQTL Mendelian randomization re-reads the outcome GWAS: an audit across six diseases and two exposure resources

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Abstract

Background. Context-specific eQTLs with Mendelian randomization are widely used to nominate immune targets in cancer. With one instrument and first-order standard errors the Wald statistic reduces to $|z| = |\beta_{\text{out}}|/\text{se}_{\text{out}}$: the exposure sets the sign, the scale and which variants are eligible, and the outcome GWAS supplies the whole significance claim. We asked what such a list contains.

Results. Instrumenting CD4⁺ T-cell cis-eQTLs from eight activation profiles against a 12,530-case melanoma meta-analysis, we confirmed the identity numerically. Benjamini–Hochberg at 0.05 corresponds to an outcome P of 2.1×10^{-4} . Significant signal concentrates on loci already reported for the outcome, 4.96-fold by bounded locus ($P = 0.0048$) — but that enrichment is carried entirely by the three loci the outcome GWAS had already found unaided at 5×10^{-8} , two over MC1R and one at PARP1. Remove them and five remain, one known: 1.98-fold, $P = 0.41$ (2.07-fold with the background restricted the same way), and the two MC1R loci are one published region, not two. Scored against their own reference lists, all five diseases with a usable list show own-list enrichment surviving Holm correction, but only melanoma and lung meet the full pre-specified criterion: colorectal is inconclusive on locus count, breast and prostate void on their mismatched controls. Colocalisation was poor here, 2 of 284 records above PP.H4 0.8,

which is consistent with outcome-dominated selection without being implied by it. Naming the gene fails separately: at ten melanoma loci with an accepted causal gene the pipeline names six, and at MC1R spans sixteen genes without naming MC1R. Replacing the outcome with a higher-powered meta-analysis replaced the list entirely, two lists from identical exposure data sharing no genes. In a 46-paper two-coder subsample of 152 comparable studies, no paper performed the locus-attribution check (0% [0–3.4]); automated matching for it had a measured precision of 0.00.

Conclusions. The reproducible part of such a list is the part that is not a discovery, and the part that would be a discovery carries no locus-level evidence we could detect — an absence of evidence, at five loci, rather than evidence of absence. Nine inexpensive checks follow, with an R implementation.

Keywords Mendelian randomization · context-specific eQTL · target nomination · colocalisation · statistical power · reproducibility · melanoma

Background

Most cis-eQTLs are context dependent, and loci mapped in resting bulk tissue cannot capture regulatory variants that act only while a cell responds to a stimulus. Datasets profiling primary immune cells across a stimulation time course were built to close that gap, and their combination with MR has become an attractive route to target nomination: genotype is fixed before disease, cis-eQTL effects carry a directional prior, and the exposure is measured in the cell type through which the effect is proposed to act. Several recent studies apply exactly this design — activation-time-course CD4⁺ T cell eQTLs as instruments, cancer GWAS as outcomes — and report novel immune targets on that basis [1–3].

The inference passes through more joints than its summary statistics reveal. Because the instrument is usually a single variant, the P value is supplied entirely by the outcome GWAS. Whether the eQTL and the disease signal share a causal variant, rather than lying in linkage disequilibrium with a large neighbouring effect, is decided by colocalisation, whose resolution depends on outcome power. Which of two genes at a locus carries the effect is not decided by MR at all; neither is the cell type in which it operates, because functional replication is usually available only at tissue level, where a broadly expressed gene reports on whichever compartment dominates. Each joint fails invisibly unless specifically tested.

Melanoma is an informative setting for two reasons that pull in opposite directions. Its common-variant architecture is dominated by pigmentation and naevus loci with effects far larger than anything expected from immune regulation — MC1R alone exceeds $P = 10^{-59}$ — which is simultaneously a built-in end-to-end positive control and a built-in confounder, since long-range LD around such loci can present as a causal association for any neighbouring gene. It is also where CD4⁺ T cell biology has the most direct clinical relevance, so candidates can be examined against response-stratified single-cell data rather than annotation alone.

The organising question of this audit is not whether any particular gene is a target, but what the nomination is a function of.

Results

Part 1 — Why the nomination is unstable

The first four results establish what a candidate list from this framework is a function of, and how much of it survives a change in the outcome GWAS.

68 **Design and positive controls**

69 Eight CD4⁺ T cell activation profiles (naive and memory, 0 h to 5 d) provided exposures.
 70 Instruments were selected at eQTL $P < 5 \times 10^{-8}$ with $F > 10$, one variant per exposure, giving
 71 3,556 records after harmonisation with the outcome. Every pre-specified control passed.
 72 Recovery of all 157 known melanoma loci, 136 of them strengthened, with genome-wide
 73 significant variants rising from 2,871 to 4,552, is an **integrity check on the outcome data and**
 74 **the meta-analysis step** rather than a positive control on the nomination pipeline — it involves
 75 no instrument, no MR and no colocalisation. The end-to-end controls are that PARP1 reproduced
 76 its published direction and that the MC1R region gave the strongest associations in the study, at
 77 $P = 4 \times 10^{-37}$ (Fig. 1).
 78 One property of the design governs everything that follows. With one instrument per exposure,
 79 the Wald z equals $\beta_{\text{out}}/\text{se}_{\text{out}}$, so **given the instrument set**, MR significance is a property of
 80 the outcome GWAS; the exposure data determine which variants are asked about and how many
 81 there are, but not which of them answers. Steiger filtering cannot arbitrate direction here either,
 82 since cis-eQTL and disease R^2 differ by orders of magnitude and the test passes almost by
 83 construction. We therefore state at the outset what the design can and cannot decide, and test the
 84 consequences empirically rather than asserting them.

85 **The nomination is the outcome GWAS, read at a lower threshold**

86 With one instrument per exposure the Wald statistic is $b = \beta_{\text{out}}/\beta_{\text{exp}}$ with $\text{SE} = \text{se}_{\text{out}}/|$
 87 $\beta_{\text{exp}}|$, so $|z| = |\beta_{\text{out}}/\text{se}_{\text{out}}|$. The exposure sets the sign and the scale of the effect estimate and
 88 the set of variants eligible to be tested; it does not enter the test statistic. We verified this on the
 89 two tables that store both sides: se matches $\text{se}_{\text{out}}/|\beta_{\text{exp}}|$ to within 0, and the reported P
 90 matches $2\Phi(-|z_{\text{out}}|)$ to within 3×10^{-16} , while the sign differs on 51% of records

(Supplementary S42). **A candidate list from this design is therefore the outcome GWAS restricted to variants that happen to be lead cis-eQTLs, thresholded by the multiple-testing burden rather than by 5×10^{-8} .** Everything below follows from that, and we report it as the mechanism rather than as a caveat.

The threshold is the first consequence. Benjamini–Hochberg at 0.05 over the 3,556 records corresponds to an outcome P of 2.1×10^{-4} — three and a half orders of magnitude more permissive than genome-wide significance.

The second is what the resulting list contains. In the FinnGen round, all ten FDR-significant records fell within 1 Mb of a known pigmentation or naevus locus — VPS9D1-AS1 50 kb from MC1R, CDK10 48 kb from the MC1R R151C variant, PARP1 5–14 kb from PARP1 — with no immune signal at FDR < 0.05 at all. Under the meta outcome, counting bounded loci rather than gene records, 4 of 8 significant loci were known-locus loci: a 4.96-fold enrichment over the 10.1% background (one-sided P = 0.0048; Fig. 1). The same enrichment counted at the record level, which is the unit the source literature reports, is Supplementary S10.

That enrichment is carried entirely by loci the outcome GWAS had already found without any exposure data. Three of the eight reach 5×10^{-8} unaided — two at 16q24.3 over MC1R and one at PARP1 — and all three are on the published list, which is how they came to be published. The cross-tabulation is completely separated: every genome-wide significant locus is a known locus and no genome-wide significant locus is novel. Removing those three leaves five loci, one of them known: 1.98-fold, one-sided P = 0.41, and 2.07-fold, P = 0.40 when the background is restricted to sub-threshold loci as well, which is the conditional comparison the question implies (Supplementary S42). **What this design adds beyond genome-wide significance shows no**

locus-level attribution signal at all — though at five loci it is also underpowered to show one,
 so this is an absence of evidence and we do not read it as evidence of absence.
 Two of the eight bounded loci are not two published regions. The windows at 16:88.86–89.73
 Mb and 16:89.87 Mb fall within 1 Mb of an identical set of seven Landi lead variants, so the
 fixed-anchor partition counts the MC1R region twice; the three genome-wide-significant loci are
 two distinct published regions, not three. Merging bounded loci that share an attributed lead
 variant, and applying the same rule to the background, gives seven regions of which three are
 known against a background of 52 of 641: 5.28-fold, $P = 0.014$ (Supplementary S42). The
 direction is unchanged and the evidence is weaker than the unmerged count suggests, which is
 the honest reading of a partition that bounds window width without guaranteeing that adjacent
 windows are independent. The same decomposition is consistent with the colocalisation result
 reported below, where 2 of 284 records reach $PP.H4 > 0.8$. It does not imply it: a variant that
 drives both expression and disease can colocalise well however the significance arose, so the low
 rate is an empirical finding compatible with outcome-dominated selection rather than a
 consequence of it.
 Loci here are the non-recursive fixed-anchor partition fixed in Supplementary S37 — a 1 Mb
 window claimed from each anchor, so a locus spans at most 1 Mb whatever the variant density;
 the single-linkage rule used in the source literature chains a dense resource into blocks of tens of
 megabases and is reported only where we reproduce published numbers (Methods). Applied
 unchanged to five other cancers, each scored against its own known-locus list rather than against
 melanoma's (Supplementary S41), the attribution recurs: own-list enrichment survives Holm
 correction across all five outcomes that have a usable list — melanoma 4.96-fold ($P = 0.0097$
 corrected), lung 3.85-fold (0.0138), colorectal 3.15-fold (0.0029), breast 2.25-fold (0.0011) and

prostate 2.04-fold (2.3×10^{-5}); pancreas has no list that clears the size and build-rate gates and is not scored. The five outcomes share almost no significant loci with one another (melanoma and lung share none; 5 of 69 distinct loci are significant in more than one outcome). Two qualifications bound this. The first is that folds are not comparable across diseases at all. Fold divides by the background known share, so its ceiling is the reciprocal of that share, and across our lists the share runs from 0.101 for melanoma to 0.403 for prostate — ceilings of 9.92 and 2.48. Matching the diseases on power by scaling each until it yields the same number of significant loci, and scoring them with the chance-corrected share $A = (p_{\text{sig}} - p_{\text{bg}})/(1 - p_{\text{bg}})$, the two statistics rank them almost in reverse: at ten loci, fold gives melanoma 5.95, lung 3.85, breast 3.10 and colorectal 2.52, while A gives breast 0.859, colorectal 0.707, melanoma 0.555 and lung 0.330 (Supplementary S43). **Melanoma's apparent lead is a property of its reference list being sparse, not of melanoma.** The second is that only melanoma exceeds every mismatched list by a margin (2.30-fold against the best rival, against 1.29 for lung and below 1 for the rest), while lung's result does not survive dropping its single strongest locus ($P = 0.0138$ to 0.0569 without the chr11 FADS1/TMEM258 cluster, the same cluster identified as the driver of lung's enrichment in our earlier melanoma-list comparison). We therefore report transport as recurring but read it as confirmatory only for melanoma.

We then tested the same attribution three further ways, changing one thing at a time (Fig. 2).

Five nested power levels of one resource

FinnGen's sequential releases carry an identical melanoma endpoint at 2,705, 2,993, 3,194, 3,932 and 5,753 cases, with exposure data, instruments, harmonisation, estimator, testing family and locus definition held fixed. Predictions were generated by down-sampling the largest release and registered before any earlier release was examined; all twelve — discovery counts, overall

recovery, and the known-minus-novel differential at each release — fell inside the registered intervals. The known-locus part of the list is recovered early and then does not move: the same five genes constitute the significant list at three consecutive releases without change, four already present at 47% of reference power. And **no novel-locus gene reaches FDR < 0.05 in any release**; the per-release lists are in Supplementary S11. Because releases are nested, agreement between them exceeds that between independent studies of equal size, so this is an upper bound on stability.

That series stops where it does for a reason. The release after it, R13, retired the endpoint we had used and replaced it with one whose cases are defined by the same codes but whose controls are screened by a different exclusion rule, so it cannot be a sixth power level; treating it as one would score a change of phenotype definition as a change in power. Analysed instead as a transfer test — pre-registered, with the direction of every difference recorded before the result was seen — the list survives it unchanged: at 6,226 cases the significant list is the same six genes at the same two loci, still with no novel-locus gene, and locus attribution is 10.0-fold ($P = 9.1 \times 10^{-5}$), all four significant loci known. The stability is not an artefact of reusing data: the two releases' statistics correlate at $|z| r = 0.937$ rather than being identical, standard errors shrink by a median 3.85% against the 3.8% the added cases predict — the internal check that R13 is the higher-powered computation and not a copy — and 51 of the 275 nominally significant records are replaced. What holds still is the significant tier; the tier below it churns. Because the extra cases and the broadened control exclusion both push towards more signal, a successful transfer here is partly confounded and we do not read it as power alone.

180 *A second disease*

181 Repeating the nomination against hepatocellular carcinoma with the exposure side unaltered, at
 182 two outcome power levels (3,748 and 947 cases), gives four significant loci across both levels,
 183 three carrying a known HCC lead SNP: SAMM50, 26 kb from rs2294915 in the *PNPLA3* region,
 184 and ZNF506 in the chr19 *TM6SF2* region at both levels. Enrichment is 15.5-fold in the lower-
 185 powered outcome ($P = 0.0041$) and 7.9-fold in the higher-powered one, where it does not reach
 186 significance ($P = 0.123$). The single novel nomination, SUPV3L1, appears only at high power
 187 and has $FDR = 0.98$ at low power.

188 Ten of that reference list's 73 loci were taken from the higher-powered outcome publication's
 189 own table, which is circular to that extent. Checked individually against the GWAS Catalog, 7
 190 carry heavy independent support (APOE in 1,399 studies, *TM6SF2* E167K in 1,073) and **3 rest**
 191 **on the outcome publication alone**; removing those three leaves both enrichments unchanged to
 192 the same decimals, because no instrument lies within 1 Mb of any of them. We report that as a
 193 measurement of the circularity — 3 of 73, none near a significant locus — not as a robustness
 194 check, since by construction it could not have moved the result. Dropping all 10 instead *raises*
 195 the enrichment, as removing known loci only shrinks the background, so over-correction flatters
 196 this test; and the Catalog carries no cohort-level information, so this addresses circular locus
 197 attribution and not sample overlap (Supplementary S29). The melanoma reference list comes
 198 from a separate publication and does not have this structure, though sample overlap with the
 199 outcome meta-analysis cannot be excluded.

200 The higher-powered HCC study carries 30.3% of melanoma's effective sample size, so we down-
 201 sampled melanoma to match: the predicted median is 2 significant loci [5–95%: 1–6], one known
 202 and one novel, and **the observed HCC counts are compatible with that melanoma-derived**
 203 **prediction interval**. The non-significant primary test is what a test with two significant loci is

expected to deliver. Compatibility with an interval spanning 1 to 6 loci is not equivalence — that interval would also accommodate outcomes we would have read as a difference — so this shows the HCC result is not evidence against the pattern, not that the two diseases behave alike once power is equalised.

A second exposure resource

Holding the melanoma outcome fixed byte-for-byte and replacing the exposure entirely with eQTLGen whole-blood cis-eQTLs ($n = 31,684$, roughly 300-fold larger, different cells, different platform) raises instruments from 3,556 to 12,835 records and significant loci from 7 to 30. The attribution does not move: **23 of 34 significant loci (67.6%) carry a known melanoma, naevus or pigmentation lead SNP, 6.31-fold over this resource's own 10.7% background ($P = 1.3 \times 10^{-15}$)**, against 4.96-fold in CD4⁺ T cells. A matched-background version agrees, drawing each significant locus a background locus of the same eQTL-p decile and allele-frequency quintile (6.49-fold, empirical $P \leq 1 \times 10^{-4}$, the floor at 10,000 permutations; 5.66-fold on the decile alone). The two resources' novel nominations intersect in exactly one gene, ZFYVE19, so the stronger claim that novel nominations never reproduce is not supported.

Three further exposure resources, and why they settle nothing

We tried three more CD4⁺ resources — Nathan et al. [4], Schmiedel et al. [5] and the influenza-stimulation panel of Randolph et al. [6] (eQTL Catalogue QTD000588). None settles the question, and the identity says why: the exposure fixes which variants are eligible for testing, not which of them are significant, so a resource with more instruments enlarges the tested set without changing what makes an entry cross the threshold. Two returned too few harmonised records to score, and the third fails its mismatched control at the registered setting. The per-resource counts and gates are in Supplementary S35.

227 ***The whole-blood rheumatoid arthritis cell is void***

228 Scored on the same records, it reaches 3.94-fold ($P = 3.3 \times 10^{-17}$), but its pre-registered
 229 mismatched-list control also enriches — melanoma's known loci give 2.05-fold on the RA
 230 nominations, $P = 2.9 \times 10^{-4}$ — and S33 §5 voids a cell whose mismatched list enriches. It stays
 231 void: excluding the MHC does not clean it (1.83-fold, $P = 0.0052$), and the 500 kb partition is a
 232 pre-registered sensitivity, not a second chance. **The non-cancer generalisation therefore rests**
 233 **on the CD4⁺ cell alone.**

234 **The control that clears this cell is clean at exactly one setting.** We swept both 1 Mb
 235 conventions — the window within which a locus counts as known, and the width of a locus —
 236 over 100, 250, 500 and 1000 kb (Supplementary S36). Across the seven distinct settings, the RA
 237 CD4⁺ cell's mismatched-list control fails at six and is clean only at the registered 1 Mb main
 238 analysis. It fails in a direction that matters: as the known-locus window narrows, the wrong
 239 disease's list enriches **more**, monotonically — 1.71-, 2.38-, 3.03- and 4.17-fold at 1000, 500, 250
 240 and 100 kb. So the control does not pass at 1 Mb because attribution there is specific; it passes
 241 because 1 Mb is coarse enough that both reference lists hit a great deal and the ratio is diluted
 242 towards one. **The cell clears its negative control at the setting where that control has least**
 243 **resolution.**

244 Two observations locate the cause. First, the contrast is within one exposure resource: melanoma
 245 and RA are scored on the same Soskic instruments, the same partition and the same convention,
 246 and the melanoma cell's mismatch enrichment is 0.00-fold at all eight settings while the RA cell's
 247 fails at six. The difference is the disease pair, not the machinery — RA and melanoma share
 248 immune loci, melanoma and hepatocellular carcinoma do not. We note against ourselves that the
 249 fragility is not confined to RA: melanoma on whole blood traces almost the same mismatch
 250 curve (4.11-, 3.44-, 2.55-, 1.59-fold as the window widens) and also fails at the tightest setting,

so a narrow window on a dense resource strains this control generally. What is specific to RA is how far the failure extends — six settings against one. Second, a per-locus breakdown rules out the convention: of the eight loci the wrong list claims at 500 kb, none is claimed through a non-significant record; every one has a significant nomination genuinely within 1 Mb of a melanoma lead. Consistent with that, RA's hit distances are spread across the intermediate range rather than bimodal — 6 of 41 within 50 kb against 15 of 34 for melanoma on whole blood — so RA's hits sit precisely in the band that a narrowing window removes.

A mismatched list is therefore not an orthogonal negative control for this outcome pair.

The enrichment itself is not the fragile part: it is flat across locus widths (3.70- to 3.96-fold) and rises as the known-locus window narrows (3.91- to 7.30-fold), the direction every other cell moves in. It is the control that is scale-bound, not the estimate.

Which disease is nominated as the mismatch changes the verdict

A single mismatched list cannot separate specific attribution from a fortunate pick, so we scored every cell against every reference list in the study with at least 30 placeable lead SNPs belonging to a different disease — six per cell (Supplementary S39). **This analysis was designed after the results it comments on and its criterion is weaker than the registered one, so we report it as a post-hoc diagnostic and draw no confirmatory conclusion from it.** Its one informative result is about the design rather than about RA: of the six unrelated lists, only one leaves the RA CD4⁺ cell standing — melanoma, the list this study designated, at 1.71-fold ($P = 0.10$) — while **hepatocellular carcinoma's at 3.05-fold ($P = 0.0028$), lung's at 3.32-fold ($P = 1.5 \times 10^{-5}$), colorectal's at 1.71-fold ($P = 0.0018$), breast's at 1.69-fold ($P = 0.0042$) and prostate's at 1.39-fold ($P = 0.026$) would each have voided it.** Whether this cell survives its negative control

therefore depends on which comparator was nominated, which is a property of single-list controls in general and one we would apply to any study using them.

The ranking that falls out is descriptive. Taking each cell's own fold against the largest fold from any of its six comparators, the six cancer cells span 2.31- to 5.52-fold while **the two RA cells sit at 1.18- and 1.32-fold, the narrowest in the grid**; RA on CD4⁺ T cells also inverts at the narrowest locus width, the only cell at any of 64 settings that does.

We therefore treat the non-cancer cell as exploratory. Its enrichment is a real observation on its own reference list, and the numbers and the full sweeps are given here and in Supplementary S36 and S39; but a cell that clears its registered negative control at one of seven settings, whose survival depends on which comparator was named, and whose margin over unrelated diseases is the smallest present, does not carry the same weight as the tumour cells. **The confirmatory claim in this paper is that the attribution appears in melanoma and in hepatocellular carcinoma. Whether it extends beyond cancer is open.**

Two further limits. Okada 2014 predates a decade of later RA loci, which count as novel here and bias the 3.91-fold downward; for the same reason enrichment magnitudes are **not comparable across diseases**, each reference list differing in age and completeness. The strongest supported statement is correspondingly narrow. Within cancer, enrichment on known outcome loci is not confined to one tumour or one exposure resource, and its negative controls are clean at every setting we swept. Outside cancer we can say that the same concentration appears in an autoimmune outcome whose pathogenic compartment is the exposure cell type, but **not** that it is specific to that outcome's own genetics, because the control which would establish that does not hold away from the registered window. Generality beyond cancer is therefore open, not established.

296 **Both axes crossed**

297 The main grid is the three diseases at their higher-powered outcome crossed with the two
 298 exposure resources — six cells — with the lower-powered hepatocellular outcome held back as a
 299 power sensitivity. Scored against each disease's own known loci, one cell is void on its
 300 mismatched control and the other five enrich:

Cell	Own fold	One-sided P
melanoma × CD4 ⁺	4.96	0.0048
melanoma × whole blood	6.31	1.3×10^{-15}
HCC-high × CD4 ⁺	7.88	0.123
HCC-high × whole blood	10.87	0.0015
RA × CD4 ⁺	3.91	2.3×10^{-7}
RA × whole blood	—	void on its mismatched control

301
 302 Four of the five reach nominal significance and remain so after Benjamini–Hochberg across the
 303 five. **That is not the relevant caveat:** the cells share exposure data and reference lists, so they
 304 are consistency of direction rather than five independent replications. The full cell-by-cell
 305 account, the registration history and the power sensitivity are in Supplementary S48.
 306 Two permutations close the gap from different sides, because a locus dense in records is not the
 307 same thing as a locus dense in a way that is itself disease-specific. Matching background loci on
 308 eQTL-p decile and allele-frequency quintile controls instrument strength and frequency: 6.66-
 309 fold, empirical P = 0.0016 for melanoma on CD4⁺ T cells. Matching on density — record count,
 310 span and gene count, requiring complete matched coverage and reporting the whole tolerance
 311 scan, with the rules fixed in Supplementary S38 before the run — gives **4.87-fold (P = 0.0050)**
 312 for melanoma on CD4⁺ T cells, 6.14-fold on whole blood, 8.24-fold for HCC-high on whole
 313 blood, 12.85- and 13.53-fold for the two HCC-low cells, and **4.52-fold for RA on CD4⁺ T cells.**

314 Values quoted at $\leq 1 \times 10^{-4}$ sit at the resolution floor of 10,000 permutations and are not point
315 estimates.

316 On bounded loci, density matching barely attenuates: melanoma on CD4⁺ T cells is 4.96-fold
317 unmatched and 4.87-fold density-matched. **That is a statement about one specified null, not**
318 **about density in general** — the 3.44-fold reported earlier came from quantile-stratified
319 matching, a different test we do not treat as the same quantity (Supplementary S38 §4).

320 **MR and colocalisation moved in opposite directions when the outcome was replaced**

321 Replacing the FinnGen outcome with the meta-analysis changed outcome power and study
322 composition together, so what follows is an observation about that replacement and not a
323 measurement of power. Meta-analysis raised MR discoveries from 10 to 21 records while
324 *lowering* colocalisation support: across the 127 exposures run in both rounds, median PP.H3+H4
325 fell from 0.249 to 0.207 and median PP.H4 from 0.124 to 0.095, with only 44.1% improving.
326 Three explanations were tested and rejected, including winner's-curse selection, for which newly
327 entering candidates had *lower* PP.H3+H4 (0.125 versus 0.202).

328 The movement is not an artefact of analysis window or prior. Across four windows (± 100 kb to
329 the full cis region) and four values of the shared-causal-variant prior, the negative results are
330 unchanged in all 16 combinations; window size is essentially irrelevant, and what varies is the
331 prior, and only for loci near the criterion boundary — TPI1 and SPSB2 pass in 8 of 16
332 combinations, SMC2 and KIAA0040 in 12 of 16. For PARP1, five orders of magnitude more
333 signal sharpened the peak and moved it 25 kb, to 54 kb from the eQTL peak, and PP.H4 fell
334 from 0.92 to 0.05; in the other direction, the meta peak for ZFYVE19 landed exactly on the
335 eQTL peak and PP.H4 rose to 0.99. One general fact belongs with this: FinnGen fine-maps only
336 19 regions genome-wide for this endpoint, all classical pigmentation or melanoma loci, and none

of our candidate loci is among them. At this outcome power the candidate list is built almost entirely on regions the outcome GWAS cannot fine-map.

The candidate list turned over completely. The FinnGen round produced four novel-locus genes passing triple validation (PRPSAP2, IMPA1, GCC2, PADI4) and one passing the full battery. Under the meta outcome none survived — IMPA1's MR P moved from 9.3×10^{-4} to 0.11 — and six different genes took their place. **The two lists share no genes**, from identical exposure data under an identical pipeline.

The reproducible part of the list is the part that is not a discovery

We measured candidate-list recovery as a function of outcome power by down-sampling in summary space, recomputing Wald ratios and FDR, and comparing with the full-power list (Fig. 3). The simulation is calibrated empirically — not externally, since FinnGen contributes to the meta outcome — against a separately observed lower-power round, which it reproduces (10.2 simulated versus 10 real discoveries; simulated Jaccard 0.52 [0.36, 0.73] containing the observed 0.455). Curves are reported only up to observed power.

Melanoma recovers 53% of the full-power list at 5,000 cases and 85% at 10,000, and five other cancers behave the same way. Stratifying by locus class dissolves that apparent stability: at 10% of observed power, 46.0% of known pigmentation and naevus-locus genes are recovered against **0.4%** of novel-locus genes; at 50% power, 85.8% versus 22.8%. Reaching 50% recovery requires 2,506 cases for known-locus genes and 8,771 for novel-locus genes — 3.5-fold. Studies using this framework commonly analyse 3,000–8,000 cases, where novel-locus recovery is 6–40%.

That differential is mostly a statement about effect size, and we tested how much of it survives conditioning on effect size. Candidates at novel loci sit against the detection threshold by

construction — their full-power $|z|$ all lie between 3.73 and 4.55, while known-locus candidates reach 15.99 (medians 4.23 and 11.07) — and recovery under down-sampling is a monotone function of $|z|$. **A model using $|z|$ with no class label reproduces 68–80% of the gap**; matched on $|z|$, the residual is **+5.2 percentage points [−1.6, +12.0] by bounded locus** and +6.4 [+0.9, +12.0] by gene, against raw gaps of 47.6 and 63.0. Only one known locus falls inside the novel $|z|$ range, so the remaining 20–32% is not separable from the failure of matching and we do not attribute it. What we retain is threshold proximity rather than a property of the category — and the practical consequence is the same either way: **for anyone holding such a list, the novel-locus part is the part that will not replicate** (Supplementary S28).

> In this analysis, and within the range of outcome power we could observe, the > reproducible part of the candidate list was precisely the part that did not > constitute a discovery.

Part 2 — Why a significant locus is not a target gene

Even where the signal is stable, the step from a significant locus to a named gene fails independently, and fails in ways the significance test cannot register.

Colocalisation and SMR/HEIDI disagree, and the disagreement is consequential

Colocalisation assigned the MC1R-region signals — the strongest MR associations in the study — to distinct causal variants, with PP.H3 dominant and PP.H4 at or near zero (CHMP1A 0.99; VPS9D1-AS1 0.96; SPATA33 0.92–1.00). FinnGen's own in-sample fine-mapping of this endpoint independently resolves three high-purity credible sets in the region ($\log_{10}\text{BF}$ 45.6, 36.6, 19.5), so it genuinely carries multiple independent causal signals: precisely the configuration in which a neighbouring gene's eQTL can be tagged by one of them without sharing it.

SMR/HEIDI did not reproduce this. Of 291 records that colocalisation assigned to distinct causal variants, HEIDI failed to reject homogeneity for 253 (86.9%), including VPS9D1-AS1

(P_HEIDI = 0.649) and CDK10 (0.086, 0.081), on 8–20 variants per test (Fig. 4). Evidence tiers that accept MR plus SMR without colocalisation would have reported MC1R linkage spillover as CD4-mediated immune targets.

Does the pipeline attribute the right gene where the answer is known?

The criticism this paper makes of nomination applies to this paper, and can be administered to it.

At melanoma loci where a causal gene is generally accepted, we asked whether the pipeline's own significant nomination names that gene. Of ten such loci reached under the larger exposure resource, **the accepted gene is among those named at six and absent at four — and is named alone at only two**, and the failures are the informative ones. At the MC1R region — the strongest melanoma locus in the genome — the nomination spans **sixteen genes and MC1R is not among them**. At the OCA2/HERC2 locus it names the pseudogene HERC2P9 rather than HERC2. At *TYR* it names ODF3, and at the CDKN2A/MTAP locus C9orf66. Where it succeeds it often does so cleanly: IRF4 and MX2 are each named alone and correctly.

Two limits bound how far this can be pushed. The accepted-gene list is one we assembled ourselves — fixed before the comparison was run, but **not pre-registered** — and ten loci is a small denominator, so **six of ten is an illustration and not an error rate**. What the check does establish is directional and does not need precision: a framework that misassigns the gene at the best-characterised locus in its own disease should not be read as assigning genes at uncharacterised ones.

Turning it into a rate would need an external, pre-defined benchmark, and we pre-registered the attempt (Supplementary S34) with a floor of 30 evaluable loci below which no rate would be reported. It could not be met: scoring our 30 significant loci against 2,435 curated GWAS locus-to-gene assignments, restricted in advance to evidence classes that do not themselves rest on

406 molecular-QTL colocalisation, leaves **16 evaluable loci and only 2 when the benchmark is**
407 **restricted to melanoma and related traits.** The binding constraint is coverage, not circularity
408 — the class restriction costs two loci. For a disease whose architecture is dominated by
409 pigmentation and naevus loci, curated locus-to-gene truth barely exists, and three-quarters of
410 what exists here comes from drug-target pairs, which follow therapeutic attention rather than
411 allelic architecture.
412 This is the co-regulation problem of Tambets et al. [7] observed at the top of the effect-size
413 distribution, and it is why we report compartment attribution separately from gene attribution —
414 the two fail independently.

415 **Which unit the significance belongs to**

416 Recomputing the significant list over record, variant, gene and bounded locus changes its length:
417 a shorter list under a stricter unit is not evidence that FDR was controlled under dependence, and
418 the point is to state which unit a conclusion is in rather than to pick the flattering one. The four
419 counts, the Simes collapsing and the consequences for every number quoted here are in
420 Supplementary S50.

421 **Part 3 — One nomination, followed as far as it goes**

422 The identity says a nomination list is the outcome GWAS at a lower threshold. It does not say
423 what happens when someone takes a single entry from such a list and tries to substantiate it,
424 which is what the field actually does. We did that once, for the one glycolytic gene this pipeline
425 nominates, and report it as a worked example subordinate to the audit rather than as a second
426 finding.

427 **It fails at every tier, and it fails in the way the identity predicts.** The gene is nominal, not
428 resolved genetically; its apparent time specificity is a statement about precision rather than

amplitude; the axis it marks is not specific to it; the tissue in which the signal is largest is not the tissue the instrument came from; the functional contrasts dissolve into lineage composition once that is matched; and the patient-level result appears in each cohort and replicates in none twice. The six rungs are recorded in full in Supplementary S47 and summarised here. Of 28 glycolytic genes, 3 are instrumentable in this exposure resource, 2 remain analysable against this outcome after harmonisation, and 1 yields a nominal association; the full gene-by-profile availability matrix is Supplementary S13 (Fig. 5). For that gene, TPI1, a usable instrument exists at 16 h and at no other timepoint, but the effect estimate is largest at rest and merely noisier there, so the window is a statement about precision rather than amplitude (Fig. 6). The axis the gene marks is enriched 21.3-fold for glycolysis and is not specific to TPI1, which is what a leave-one-out reconstruction with the gene removed from both the defining score and the enrichment family shows (Supplementary S31; Fig. 7). In melanoma single-cell data TPI1 is 2.75 log₂ units higher in malignant cells than in the CD4⁺ T cells the instrument came from, in 16 of 16 patients; the compartment attribution across cell-type, within-patient paired, spatial and bulk data is Fig. 8. Splitting cells by glycolytic score splits them by lineage purity, and the helper and regulatory differences collapse once lineage composition is matched at cell level. In patients, both arms are significant in the discovery cohort and the result appears in each of the three cohorts without replicating in any two of them; the locked signature is Supplementary S17 and the inference, including all three treatments of repeated patients, Supplementary S19 (Fig. 9).

Nothing on that ladder makes TPI1 a target, and we do not present it as one. Its value here is the price it puts on each rung of a ladder the identity predicts is not climbable, and every one of those prices was paid before we understood why.

Part 4 — These checks are rarely reported

From 469 screened records, 209 were eligible and 154 had an open-access full text, of which 152 were scorable. In that sample, **not one paper in a random 30% subsample compared its significant signal against previously reported loci for its own outcome trait (0% [0–3.4]), and an estimated 7.1% [2.5–16.1] report how the candidate list depends on the outcome GWAS used.** About 59% performed colocalisation and 47% used SMR or HEIDI. Three studies using this framework closely enough to compare item by item are set beside ours in Supplementary S16.

Those are corrected figures. Automated matching returned 7.9% and 35.5%; **two coders independently scored a random 30% subsample (46 papers, 92 judgements), blind to each other and to the automated score**, and its precision proved to be 0.00 and 0.20. Most hits were spurious — three of five on the first criterion lay inside reference lists, where a cited GWAS title containing "novel loci" matches, and twelve of fifteen on the second were generic statements about statistical power. **Neither of our published numbers survived:** 7.9% was too generous, and after adjudication no paper in the subsample performed that check at all, while 0.7% for the second criterion was too strict and falls below the corrected interval. We had reported the permissive count for one criterion and the strict count for the other.

Agreement was 97.8% on each criterion with one disagreement each; Cohen's κ was 0.79 for list stability and **0.00 for locus attribution — a prevalence artefact, since with one positive in 46 the expected agreement equals the observed**, which is why the raw agreement is given beside it. Both disagreements were adjudicated jointly and both resolved against the first coder; κ is computed before that. Recall was bounded from above rather than measured, and a full-text audit measures reporting, not practice (Methods).

474 How far the identity reaches into this literature is a question about the literature rather than about
 475 the algebra, and our first attempt to answer it failed. Scanning the 154 cached full texts — two
 476 more than the 152 scorable ones, because scorability required the five pre-fixed criteria to be
 477 assessable and the scan needs only the text — 71 contain a phrase asserting a single-variant
 478 instrument and 130 name a multi-instrument estimator. **Neither count survives inspection.** 23
 479 of the 71 never mention a cis-eQTL anywhere; requiring the phrase to fall within 500 characters
 480 of one leaves 13, and within the same sentence, 10. **A count that falls from 71 to 10 under a**
 481 **minimal proximity requirement is a phrase-hit tally of unknown direction, not a bound in**
 482 **either direction, and we withdraw it as one.** The quantity the question needs is the estimator
 483 that produced each paper's primary cis nomination list, which phrase matching cannot see at all:
 484 naming IVW does not establish that the cis list came from IVW, and saying "single variant"
 485 somewhere does not establish that it came from a Wald ratio. We are coding that provenance by
 486 hand, as a sixth criterion on the same 46-paper two-coder subsample, under rules fixed before
 487 any paper was read (Supplementary S51). Until it returns, **how much of this literature the**
 488 **identity reaches is unmeasured**, and the scan is reported as a corpus-discovery tool rather than
 489 as a result.

490 Discussion

491 The results have a single mechanism, and it is algebraic rather than empirical. With one
 492 instrument and first-order standard errors the Wald statistic is $|\beta_{\text{out}}|/\text{se}_{\text{out}}$, so a candidate list
 493 from this design is the outcome GWAS restricted to lead cis-eQTLs and thresholded at whatever
 494 the multiple-testing burden allows — here 2.1×10^{-4} . One consequence is algebraic and the rest
 495 are not, and the distinction matters. What follows algebraically is narrow: signal must land where
 496 the outcome GWAS has signal, and profile specificity cannot appear in a P value, because the P

value contains no exposure term. Everything else below is an empirical observation that the identity makes likely rather than necessary. Whether the outcome's strongest signals among lead cis-eQTLs sit on previously published loci depends on where cis-eQTLs are and on the disease's genetic architecture, not on the algebra — here three of our eight significant loci reach 5×10^{-8} unaided, all three are on the reference list, and removing them leaves 1.98-fold, $P = 0.41$ (2.07-fold, $P = 0.40$ with the background restricted to sub-threshold loci as well). The attribution recurred in every further disease we could test, which the identity did not require. Colocalisation was poor in this application, 2 of 284 above PP.H4 0.8; the identity does not require that either, since a variant that drives both expression and disease can colocalise well.

We think this reframing is the useful contribution, and we state plainly what it is not. The reduction itself is elementary and not new. It does not apply to multi-instrument estimators, where combining instruments breaks the identity — our relaxed IVW and weighted-median analyses are outside it. How much of the literature that exempts is unmeasured. We tried to bound it by phrase matching and the attempt failed, for reasons we give in full in Supplementary S51; a hand-coded estimate of which estimator produced each paper's primary cis nomination list is registered on a 46-paper subsample and has not yet returned. It does not make MR effect estimates wrong; the sign and the scale genuinely come from the exposure, and only the significance claim does not. And it does not show that the sub-threshold part of a nomination list is spurious: at five loci we are underpowered to say anything about it, which is a different statement from showing there is nothing there.

Everything in this framework that depends on the outcome GWAS proved unstable, and the instability has a specific shape: signal lands on loci the outcome already knows about, the novel part of the list turns over completely between outcomes, and replacing the outcome with a larger

and compositionally different meta-analysis moved MR and colocalisation in opposite directions. The pattern held when we changed the disease, when we changed the exposure resource by two orders of magnitude, and in five of the six cells of the two crossed — three diseases by two resources, four of them reaching nominal significance and three of those carrying confirmatory weight. In the sixth, rheumatoid arthritis on whole blood, the mismatched-locus control itself enriched, and we report the cell as void rather than as a result. The control is the load-bearing part of that sentence, and it does not carry equal weight everywhere. Three of the cancer cells show no mismatch enrichment at any window we swept; a fourth, melanoma on whole blood, fails at the tightest window only; rheumatoid arthritis on CD4⁺ T cells is clean at the registered window and fails at every tighter one. Where a negative control is scale-dependent, so is the claim it licenses, and the control is most scale-dependent exactly where the outcome pair shares biology. Scoring every cell against six unrelated lists instead of one — a post-hoc diagnostic, not a registered test — sharpens the same point into a design lesson we would apply to any study using this control: which disease is nominated as the mismatch determines the verdict. On our own non-cancer cell, two of four candidate comparators leave it standing and two would have voided it. A control with one arbitrary comparator measures the comparator. What did not depend on the outcome was a smaller and more durable statement: in which cell state a pathway's regulation can be measured precisely enough to yield an instrument at all. Of eleven target-level tests that returned a verdict, seven overturned the claim under examination, three were inconclusive and one was significant on discovery but not confirmed on transfer; five genes were successively designated lead candidate under a ranking criterion that was never fixed in advance, and the first four designations were overturned. That is this project's own history and not an estimate of how often the framework fails, and the itemised record —

every attempt, stopping rule and withdrawal, the candidate-selection timeline, the fourteen pre-registrations and the technical account of two processing errors of ours — is Supplementary S12. Eight checks follow directly, each cheap and each capable of changing what a study of this kind reports.

Eight checks follow from the failures above. Each is stated in full, with the numbers behind it, in Supplementary S49, and all nine — these eight plus the decomposition of S42 — are implemented in the `cqtna` package.

(i) Annotate the signal against the outcome's own known loci, counting. (ii) Require colocalisation, preferably with explicit multiple-signal modelling. (iii) Do not assume that more outcome power will resolve the disagreement. (iv) Report the physical distance between GWAS and eQTL peaks alongside. (v) Compute the nominated gene's cell-type expression ratio in an annotated single-cell dataset for the tissue, before any functional interpretation. (vi) Report instrument availability at each of the three levels it passes. (vii) When splitting cells by a score, match on lineage composition as well as. (viii) Before comparing candidate lists across releases of the same GWAS.

The worked example should be read in that light, and is best stated as a ladder rather than a verdict — the same layered reporting Howe et al. use when a cellular model proves only partly transportable [8]:

Claim about TPI1	Status here
Causal target for melanoma	Not supported — FDR = 0.119, prior-dependent colocalisation, region not fine-mappable
Dynamic <i>genetic</i> effect across activation	Not supported — no genotype × pseudotime interaction at any of three lead variants
Dynamic <i>expression</i> across activation	Supported — Moran's I = 0.664
Membership of a definable CD4 ⁺ metabolic state	Supported — ranks 19th of 7,653 on an axis built without it, and 18th and 28th in a second dataset with protein-based lineage calling; axis enrichment 21.0–27.4-fold

Claim about TPI1	Status here
Predicts checkpoint-blockade response	Unstable — significant on discovery, no arm confirmed twice
Functional consequence isolable by perturbation	No — a hit in 628 of 1,471 human CRISPR screens (42.7%), a core-essential profile that cannot isolate a CD4-specific role
Recognised as a target for this disease elsewhere	No — in Open Targets [9], TPI1's strongest disease associations are triosephosphate isomerase deficiency and neurodegenerative disease; melanoma is not among its leading associations

Stated flatly: TPI1 is a nominal, prior-sensitive candidate selected by a multi-layer conjunction — MR $P = 1.4 \times 10^{-3}$ but FDR = 0.119, PP.H4 = 0.51 passing in 8 of 16 window-and-prior combinations, in a region the outcome GWAS cannot fine-map, with gene attribution at chr12p13 unresolved by genetics. **What MR supplies here is a genetically tractable entry point into a pathway, and the audit establishes that the gene marks an expression state — not that the state is genetically driven, and not that the gene is a target.** Its value is as a demonstration that instrument availability and biological importance are separable: the strongest enzyme in the functional data carries no instrument, and the gene that does carry one is principally expressed by tumour. The complete record of which claims about it were raised, tested and withdrawn, together with the selection denominators at gene, pathway and mechanism level, is Supplementary S12; the fourteen pre-registration documents are enumerated in Methods. **Relation to existing guidance.** These eight are additions to, not a replacement for, current cis-MR practice. Existing guidance already stresses that cis analyses must be tailored to local biology, that an expression biomarker is not an intervention, that co-regulation of neighbouring genes can act as horizontal pleiotropy, and that MR is one strand of triangulation [10,11]. Tambets et al. showed with approximate ground truth that neighbouring-gene co-regulation routinely produces colocalisation evidence for several genes at once, so eQTL data suit candidate *generation* and evidence *composition* better than gene *attribution* [7]; our MC1R region is the

same phenomenon at higher effect size, and TPI1 at chr12p13 is the same problem unresolved. Reales et al., auditing over a million colocalisation tests across immune diseases and cell types, likewise found that gene assignment shifts with platform coverage, cell context and resource size [12]. What we add is specific to the nomination step and to the outcome side: locus attribution against the outcome's own known loci, sensitivity to the outcome release, candidate-list stability, layered instrument visibility, and cell-compartment attribution.

Limitations.

What the grid can and cannot support. Three diseases and two exposure resources, with the second hepatocellular power level drawn from the same resource as the first, show that the pattern is not peculiar to melanoma or to one small eQTL dataset — not that it holds for any disease or any resource. One of the six cells is void on its own mismatched control, one of the remaining five reaches neither $P < 0.05$ nor the density-matched null, and the one non-cancer cell that survives does so at the registered window only, its control failing at six of seven settings and failing harder as the window narrows. We therefore claim the attribution as observed in two tumours, not in tumours in general, and treat the autoimmune cell as exploratory. Three diseases is not a sample from which generality can be estimated; they were chosen for the contrasts they provide. The six comparator lists are all cancers — genuinely unrelated to rheumatoid arthritis, but not a random sample of diseases, and a stronger non-cancer claim would need comparators outside oncology and immunology and a negative control specified in advance of the cell it judges.

What the diagnostic can and cannot do. The mismatched-list control is informative only inside a bounded power range: as the significant set grows towards the background the fold necessarily approaches 1 and unrelated lists begin to enrich too. Across the outcomes scored against their

own lists the own-list fold falls from 4.96 at eight significant loci to 2.04 at twenty-eight, and at the upper end the comparators enrich together — for prostate and breast, whose lists share 43% of their flagged loci, own-versus-mismatch cannot discriminate by construction. A clean control at low power and a failed one at high power are not directly comparable, and the melanoma cell's clean controls are in part a property of its having only eight significant loci. The power-recovery curves rest on reference lists of four and six genes and compare against a full-power list that is itself unstable, so they measure agreement between two imperfect lists rather than recovery of truth.

What the exposure side cannot reach. Both resources carrying the attribution finding are European-ancestry and the second is whole blood, so a positive result there cannot be read back as CD4-specific; the three further resources we tried, which include Peruvian and African-ancestry donors, were underpowered or failed their control. **That limit is structural.** Across the eQTL Catalogue [13], multi-condition stimulation resources at adequate donor numbers exist only for myeloid cells, while for CD4⁺ T cells the only multi-condition resource is the one used here and every other stimulated CD4⁺ dataset carries a single condition. **The dynamic axis of the framework this paper audits rests on one resource**, which bounds how far any audit of it, including ours, can go.

What needs new data rather than new analysis. An external validation protocol is adopted and frozen (Supplementary S40), restricting the confirmatory question to the melanoma outcome-side claim in a cohort with no participant overlap with anything used here. It is not registered, no qualifying cohort has been secured, and it therefore carries no realised power and is stated as a commitment. The patient analyses rest on three small cohorts of which the strongest is the discovery cohort, and the genotype $\times$ pseudotime test covers lead variants under linear and

quadratic models, so a non-monotonic effect at a non-lead variant is not excluded. Two questions cannot be settled here at all: whether the patient stratification is real, which needs a same-disease, same-regimen, CD4-resolved cohort that does not exist publicly, and whether the state has immunological consequences, which needs perturbation. **Conclusions.** A candidate list produced by this framework at current outcome power should be reported as the set of genes that passed screening under these conditions, not as targets — and because the fragile part of such a list is precisely its novel-locus part, the headline of a study of this kind is the part least likely to replicate. The eight checks above cost little and would have changed what this analysis reported at nearly every stage.

Methods

Exposure data

Cis-eQTL summary statistics came from a published activation time-course of primary human CD4⁺ T cells [16], comprising eight profiles — naive and memory cells at 0 h, 16 h, 40 h and 5 d — from 85–100 donors each (naive 99/99/89/85; memory 100/95/89/90). Per-profile sample sizes were recovered from $\text{ma_count} / (2 \cdot \text{MAF})$ and were internally consistent for 100% of variants in all eight profiles. Expression had been inverse-normal transformed, so exposure units are standard deviations. Coordinates are GRCh38 and were used as distributed; no liftover was performed. The second exposure resource was the eQTLGen Consortium whole-blood cis-eQTL release ($n = 31,684$) [17].

Outcome GWAS

The primary outcome was a fixed-effect inverse-variance meta-analysis of FinnGen R12 C3_MELANOMA_SKIN_EXALLC (5,753 cases / 378,749 controls; the OpenGWAS finn-b-

snapshot was not used, being an R5 extract with 98 cases) and Rashkin et al. (GCST90011809, 6,777 cases) [18,19], totalling 12,530 cases and 789,099 controls ($N = 801,629$, case fraction 0.015631). Rashkin reports odds ratios and P values without standard errors, so standard errors were reconstructed as $se = |\log OR|/|\Phi^{-1}(P/2)|$. For a two-study meta-analysis Cochran's Q has one degree of freedom, so the upper tail is $\text{erfc}(\sqrt{Q/2})$; the $\exp(-Q/2)$ form used initially gave a significant-Q rate of 1.59% against 5.33% for the correct form. 1,740 variants underflowed to $P = 0$, for which log-space $mlogp$ was computed and P floored at 10^{-300} . Landi et al. (GCST010304) supplied only 76 loci and was used solely as the known-locus reference [20]. Further outcomes were FinnGen R12 lung, colorectal, pancreatic, breast and prostate cancer; FinnGen R12 C3_HEPATOCELLU_CARC_EXALLC (947 cases) [21] and GCST90809296, the European arm of an eleven-cohort hepatocellular meta-analysis (3,748 cases / 1,861,536 controls) [22]; and FinnGen R13 M13_RHEUMA (16,775 cases / 308,847 controls). Effective sample sizes, $4/(1/N_{\text{case}} + 1/N_{\text{control}})$, are 49,337 for the melanoma meta, 14,962 and 3,779 for the two hepatocellular levels, and 63,643 for rheumatoid arthritis.

Instrument selection, harmonisation and MR

Instruments were the top cis-eQTL per gene $\times$ profile at $P < 5 \times 10^{-8}$, matched to the outcome by chr:pos with allele consistency; 26 variants were discarded for ambiguous indel representation. The strict set is 3,556 records over 2,126 unique variants for the meta round, and 3,579 over 2,142 for the single-outcome round. The minimum F statistic was 36.1 in either strict set and 22.2 across the harmonised set before instrument selection, so the $F > 10$ filter was never binding at any stage. Each exposure carries one instrument, so the primary estimator is the Wald ratio with first-order delta-method standard errors. The analytic consequence is stated because it constrains interpretation: with $b = \beta_{\text{out}}/\beta_{\text{exp}}$ and $SE = se_{\text{out}}/|\beta_{\text{exp}}|$, the statistic $z =$

$\beta_{\text{out}}/\text{se}_{\text{out}}$ does not involve the exposure, so a variant returns the same P value in every profile in which it is the lead eQTL. Profile specificity is therefore reported as the profile containing the instrument, never as a difference in P value.

A relaxed set ($F \geq 5$, $P < 0.05$, LD-clumped with PLINK 2 at `--clump-r2 0.1 --clump-kb 1000` against 525 unrelated 1000 Genomes GRCh38 EUR samples [23,24]) supported IVW, multiplicative-random-effects IVW and weighted median sensitivity analyses [25]. Weighted mode was evaluated and excluded: at a median of four instruments per exposure it returned 1 significant result of 226 testable exposures. Because clumping at $r^2 < 0.1$ retains correlation while IVW assumes independence, IVW significance is inflated (157 exposures at $\text{FDR} < 0.05$ against 21 in the strict set), so sensitivity results are reported as concordance between IVW and weighted median rather than as independent discovery. Multiple testing used Benjamini–Hochberg FDR within each analysis family, the family fixed before the analysis ran. Steiger filtering used the recovered per-profile sample sizes, SD exposure units, log-odds outcome units and outcome prevalence 0.014962; all records had the correct direction. TwoSampleMR's R^2 formula for SD units is unbounded and produced values above 1 (maximum 1.027), so the bounded form $R^2 = F/(F + N - 2)$ (range 0.185–0.933) is additionally reported.

Unit of inference. The testing family is gene $\times$ profile records, and the record level is primary throughout, because it reproduces the pipeline under audit: the registered down-sampling predictions, the release trajectory, both generalisations and the grid were computed on it. Records are not independent — 3,556 carry 2,126 unique variants and 1,195 unique genes — so the list was recomputed under seven alternatives as a sensitivity analysis: variant, gene and bounded locus, each collapsed by minimum-p and by Simes combination, plus a two-stage hierarchical procedure selecting genes by Simes and then records within selected genes. All

seven return a superset of the record-level gene list. The audit's own conclusions are stated in independent loci.

Colocalisation, SMR and HEIDI

Colocalisation used `coloc.abf` with default priors over the full cis window [26], the eQTL as a quantitative trait and the disease as case-control, matching on a chr:pos:alleles key. The published criterion $PP.H4/(PP.H3+PP.H4) > 0.7$ is too permissive here, since the ratio can rest on two vanishingly small posteriors, so $PP.H3+PP.H4 > 0.5$ was additionally required. Every result carries two diagnostics: the number of variants in the region, and the physical distance between the GWAS and eQTL peaks. Sensitivity across four windows (± 100 kb to the full cis region) and four values of the shared-causal-variant prior (p_{12} from 10^{-6} to 5×10^{-5}) is reported for every locus. Because `coloc.abf` permits at most one causal variant per trait per region, the assumption was tested on the outcome side using FinnGen's published in-sample SuSiE fine-mapping and `susie_rss` run on 1 Mb windows with a proxy LD matrix from the same 525-sample panel [23,27], with `estimate_s_rss` as a mismatch diagnostic. The eQTL side was not fine-mapped: at 85–100 donors without in-sample LD, SuSiE is not reliable there, and this is a stated boundary. Our proxy-LD procedure recovers 19 credible sets at MC1R against 3 from in-sample fine-mapping, so its counts are read in one direction only — exactly one credible set is conservative evidence of a single signal, several is not evidence of multiple signals. SMR v1.3.1 [28] was run against the same reference panel (94,979 within-cis variants) with GWAS statistics trimmed to cis windows and de-duplicated for multi-allelic sites. The `--diff-freq` QC removed 0.45% of variants at the default threshold, which was not relaxed. The number of variants entering each HEIDI test is reported with the result, as HEIDI's power depends on it (range 8–20 here).

717 Locus annotation and enrichment

718 The melanoma reference set was the union of lead variants from the three Landi et al. 2020
 719 GWAS [20] retrieved from the GWAS Catalog REST API [29] and mapped to GRCh38: 157
 720 loci. Hepatocellular loci were lead variants for EFO_0000182 and liver cancer at $P < 5 \times 10^{-8}$
 721 together with those tabulated in the outcome publication — 83 rsIDs, 73 placeable on GRCh38.
 722 The rheumatoid arthritis reference was the 87 lead rsIDs of Okada et al. 2014 [30]
 723 (GCST002318), published before FinnGen existed and from cohorts that exclude it; 83 were
 724 placeable. Loci are a **non-recursive fixed-anchor partition** of instrument positions within a
 725 chromosome — bounded rather than independent, since the partition guarantees only that two
 726 loci share no variant: the first unassigned variant becomes an anchor and claims every variant
 727 within 1 Mb of it, then the next unassigned variant becomes the next anchor. Anchors are taken
 728 by position and never by significance, because the same partition supplies the numerator and the
 729 denominator, and a locus therefore spans at most 1 Mb whatever the variant density. A locus
 730 counts as known if **any** of its instruments lies within 1 Mb of a reference lead; the same
 731 convention scores background and significant loci, since a background locus has no significant
 732 record to be scored by. The partition rule, the 1 Mb main window, the 500 kb sensitivity and this
 733 convention were fixed together in Supplementary S37 before the last four cells of the grid were
 734 computed.

735 The single-linkage rule used in the source literature — variants joining transitively at 1 Mb — is
 736 retained only where we reproduce published numbers, and carries no inference here. It is not
 737 interchangeable with the above: on the whole-blood resource it chains a chromosome arm into a
 738 single 30.8 Mb "locus" of 588 records, and a block that wide contains a known lead SNP for
 739 almost any disease. Under that rule the pre-registered mismatched-list control fails on the whole-
 740 blood melanoma cell, which is what prompted the change; under the fixed-anchor rule the widest

significant locus is bounded at 1 Mb by construction and that control is clean (Supplementary S37 §2). Enrichment among FDR-significant loci was tested one-sided by Fisher's exact test against the background proportion over all instrument loci. Every reference list was built and written to disk before the MR step it scores was run.

Three controls bound this test. A **mismatched-list** control scores each disease's significant loci against another disease's reference. Because the verdict of that control depends on which disease is nominated, it is accompanied by a **multi-list** diagnostic (Supplementary S39): every cell is scored against every reference list in the study that has at least 30 placeable lead SNPs and belongs to a different disease, six per cell, and we report how many also enrich together with the ratio of the outcome's own fold to the largest fold from any comparator. **This diagnostic was specified after the results it comments on, and its criterion — that the outcome's own list exceed every comparator — is weaker than the registered rule that voids a cell whose mismatched list enriches at all. It is reported as exploratory and is not used to upgrade any cell.** The 30-lead floor is the one already used in Supplementary S34 and no new lists were obtained; the breast and lung lists were, however, rebuilt from the same GWAS Catalog rsIDs after we found that the original coordinate lookup had silently dropped whole request batches, leaving 6 and 0 placeable leads where 740 and 271 were available (Supplementary S41). Under the repaired panel the melanoma CD4⁺ cell acquires one nominally enriching comparator, breast at 2.15-fold ($P = 0.0495$), so its multi-list verdict moves from no enriching comparator to one; the cell's margin over its best rival widens from 1.82- to 2.31-fold, because the published prostate comparator had been computed from 35 of its 1,160 leads. Under Holm correction across the six comparators the breast result does not reach significance, so this verdict depends on which multiplicity rule is applied and we report both. A **strength-and- frequency**

permutation matches significant to background loci on eQTL-p decile crossed with outcome allele-frequency quintile, with 10,000 stratified resamples and a one-sided empirical P. A **density** permutation follows the estimand in Supplementary S38, whose matching rule, coverage requirement and sampling scheme were fixed in a document first committed before the run; the choice of 1.00 as the reported tolerance was recorded in the same commit as the results and is therefore an analysis choice rather than an independently timestamped one. Background loci are matched to significant loci on record count, physical span and unique-gene count simultaneously, drawn without replacement from a pool that excludes every significant locus, over 10,000 replicates with a one-sided empirical P. The matching tolerance is fixed at 1.00 and the full tolerance scan is reported alongside; a cell whose matched coverage falls short of 100% returns no P value rather than a partially matched one, and four cells do so at the tightest tolerance. An earlier quantile-stratified version (Supplementary S30) is retained but is a different test and is not pooled with this one. All three controls are computed on the same locus partition as the estimates they bound.

Circularity of the hepatocellular reference. Ten of its 73 loci came from the higher-powered outcome publication's own table. Provenance was resolved through the Catalog REST API by retrieving that study's reported associations and querying each locus for the number of other studies reporting it: seven have substantial independent support and three rest on that publication alone. Attribution was recomputed against the published list, the list without those three (primary), and the list without all ten as an over-corrected bound. Note that removing known loci shrinks the background and can only raise the enrichment, so the over-corrected version is not a stricter test.

Power trajectory, transfer test and down-sampling

Five FinnGen releases carrying an identical melanoma endpoint (R8–R12) were analysed with exposure data, instrument set, allele orientation, harmonisation, estimator, testing family and locus-merging rule held fixed, so the systematic difference between runs is case number. The primary analysis was restricted to variants present in every release. Predictions were generated by down-sampling R12 over 2,000 replicates and registered before any earlier release was examined. Because releases are nested, agreement between them exceeds that expected between independent studies of equal size, so the design is conservative with respect to instability.

R13 replaced C3_MELANOMA_SKIN_EXALLC with C3_MELANOMA_SKIN_WIDE (6,226 cases). Endpoint definitions were compared code by code in Ristey's rather than read from the release manifest, whose one-line description ("including Hilmo") had led us to the wrong conclusion at first: the two endpoints select cases with the same codes and differ in the rule excluding cancers from the controls. R13 was therefore excluded from the trajectory, whose comparability gate requires verbatim-identical endpoints, and analysed as a transfer test with everything but the outcome file held fixed. Three process controls were required before any R13 number was interpreted: exact reproduction of the R12 candidate list through the new code path, presence of the large-effect MC1R-region genes, and instrument coverage of at least 95% of the R12 set.

Power was reduced in summary space, individual-level data being unavailable: for a target effective size, $se_{sim} = se_{obs} \cdot \sqrt{(N_{obs}/N_{sim})}$ and $\beta_{sim} = \beta_{obs} + \varepsilon$ with $\varepsilon \sim N(0, se_{sim}^2 - se_{obs}^2)$. Wald ratios and BH-FDR were recomputed over 200 replicates per grid point and the $FDR < 0.05$ list compared with the full-power list by Jaccard index, sensitivity and precision. Calibration was pre-specified as the condition for reporting any curve: reducing the meta outcome to FinnGen's effective size reproduces the independently observed FinnGen round (10.2 simulated against 10 observed discoveries; simulated Jaccard 0.52 [0.36, 0.73] containing

the observed 0.455). We call this empirical rather than external calibration, because FinnGen contributes to the outcome being down-sampled. Two attempts to extrapolate above observed power were abandoned — a flat prior overpredicted discoveries at observed power 25-fold and global empirical-Bayes shrinkage underpredicted them 100-fold — both caught by the pre-specified requirement that any prior reproduce the observed discovery count. **All curves are reported only up to observed power and must not be extrapolated.**

Effect-size matching. The down-sampling model takes no class label as input, so the known-versus-novel differential can only follow from the two classes' $|z|$ distributions. Each unit's recovery frequency at half power was regressed on $\log|z|$ with and without the class label, and novel units were matched to known units within a pre-registered caliper of 0.20 on $\log|z|$, with replacement, nearest first, the residual bootstrapped over matched pairs. Common support was recorded, since the matched comparison is interpretable only where the $|z|$ distributions overlap. The locus level was pre-specified as primary. A process control required the reimplementations to reproduce the published recovery curve bitwise before any matched result was read; this failed on first run because the published script draws from one generator consumed across all power points while the reimplementations re-seeded per point, and was resolved by matching the draw sequence rather than by loosening the tolerance.

Self-administered attribution check

Of the 30 loci reaching $\text{FDR} < 0.05$ under the whole-blood exposure, 20 are known loci, and those 20 are the domain of this check. A locus's nomination is the set of FDR-significant genes there whose own instrument lies within 1 Mb of a known lead. Melanoma loci at which a causal gene is generally accepted were listed, and the nomination at each was compared against that gene under two rules fixed before the comparison: *lenient*, the accepted gene appears anywhere

in the nomination, and *strict*, the nomination is that gene alone. Ten of the 20 had an accepted gene. The list was assembled by us and fixed before the comparison was run, but was **not** pre-registered, and ten loci is a small denominator; the resulting ratios are reported as an illustration and not as an error rate.

Single-cell, spatial and chromatin analyses

Full processing details for the ICB cohorts (GSE120575, GSE115978, GSE72056, GSE78220, GSE91061, GSE235863) [31–36], the three spatial datasets [37–39], the purified CD4⁺ multiome (GSE282266) [40] and the peripheral-blood variance decomposition (GSE199994) [41] are given in Supplementary S12; the points that changed a result are summarised here. Donors or samples, never cells, are the statistical unit for inference. Module scores used Seurat `AddModuleScore` [42] with expression-binned control genes, and every module used as evidence independent of the nominated gene was recomputed with that gene removed. Because splitting cells by a score also splits them by lineage purity, every within-population split was controlled at the **cell** level — lineage-marker-free module versions, residualisation on a CD8ness score, and matching of high- and low-score cells on CD8ness and sequencing depth, with residual standardised mean differences reported for both.

The metabolic axis was defined by residualising a 16-gene glycolysis module on an eleven-gene activation module and on sequencing depth, with a pre-specified kill criterion — if $R^2(\text{glycolysis} \sim \text{activation} + \text{depth}) > 0.70$ the axis is not separable from activation and the analysis stops.

Observed $R^2 = 0.024$. Cells were matched on activation decile $\times$ depth quintile before splitting on the residual, and only genes concordant in direction across all four sets were carried forward.

A nuclear-retention index (MALAT1 + NEAT1 fraction) carried its own kill criterion, and module detection rates were checked before any module-level null was interpreted; three pre-

specified immune modules were withdrawn on that basis as unmeasurable rather than negative.

The axis was originally circular with respect to the nominated gene, which entered both the defining score and the enrichment family, and was rebuilt with that gene removed from both, everything else held fixed, leaving its rank a held-out quantity.

Replication of the axis. GSE166188 (DOGMA-seq, STIM arm, 16 h) was analysed with the construction held bitwise identical to the discovery run and only the dataset changed, pre-registered before it was run (Supplementary S32). CD4 cells were gated on CLR-normalised surface protein (CD4 above median, CD8 at or below), not on transcript. Its two arms are two lysis buffers on one batch of material — technical replicates, not two donors — so this replicates across study, platform and lineage calling and not across donors. Two dimension conventions differ between the protein and RNA matrices in this resource, and dimension assertions were added after an earlier script indexed the protein matrix by the RNA convention; the protein file also retains unfiltered empty droplets, so gating requires intersecting with the RNA barcodes first.

Chromatin peaks differ between samples, so cross-sample comparison was performed on merged genomic intervals (199,740 consensus intervals) with the global index preserved. Motif enrichment used JASPAR2020 CORE with `motifmatchr` against hg38 [43,44], comparing foreground with background peaks matched on GC content and log accessibility. Because axis and activation $\log_2$ fold-changes correlate at $r = 0.647$ at the chromatin level despite near-orthogonality at the RNA level, the analysis was repeated restricted to activation-invariant peaks; both versions are reported, at motif-family rather than individual-factor level.

877 **Literature audit**

878 Four PubMed queries returned 469 records, of which 209 were eligible and 154 had an open-
879 access full text, 152 of them scorable (Supplementary S23). Five scoring criteria were fixed
880 before scoring. Because automated matching is imprecise on this task, a fixed-seed random 30%
881 subsample (46 papers, 92 judgements) was scored by **two coders independently, each blind to**
882 **the other and to the automated score**: the handover file carried neither the first coder's
883 decisions nor the automated label, and rows were shuffled under a fixed seed so that automated
884 positives were not clustered. Measured precision was 0.00 and 0.20 on the two criteria, and the
885 field-level estimates reported in the text are precision-corrected and supersede both the generous
886 and the strict automated counts. Cohen's κ is reported beside the raw agreement, because at one
887 positive in 46 the expected agreement equals the observed and κ collapses to 0 while agreement
888 is 97.8%. Disagreements were adjudicated jointly, never by the first coder alone, and κ is
889 computed before adjudication. Recall was estimated by probing negatives for near-miss wording
890 rather than by reading all 152 in full, so it bounds sensitivity from above; and a full-text audit
891 measures reporting, not practice.

892 On the same corpus we counted, by phrase matching, whether each paper names a Wald ratio,
893 IVW, weighted median, MR-Egger or mode-based estimator and whether it states anywhere that
894 its cis instrument is a single variant. Those counts are reported as corpus description and not as
895 an estimate of anything: the single-variant phrase is matched anywhere in the text, so it is not
896 tied to the cis analysis, to the primary candidate list or to the estimator used, and the count
897 collapses from 71 to 10 when the phrase is required to appear in the same sentence as a cis-eQTL
898 mention. The interpretable quantity is which estimator produced each paper's primary cis
899 nomination list. It is being coded by hand as a sixth criterion, C6, on the fixed-seed 46-paper
900 subsample already used for the other criteria, by two coders independently, each blind to the

other and to the automated scan, under rules fixed before any paper was read, with the evidence passage recorded for every judgement; agreement, Cohen's κ , the adjudicated values and an interval will be reported with it (Supplementary S51).

Pre-specification and stopping rules

The following were fixed before the analyses they govern and are reported because they determined which results are presented. Every test carries a positive control, and a test whose positive control fails is discarded rather than interpreted — the within-lymphoid-subset spatial test (HLA-C $P = 0.23$) and the pre-specified immune-module analysis were discarded under this rule. Kill criteria, test direction and window sizes were written into each script before it ran. A negative result was not followed by a search for a positive one in the same data. Simulations require calibration against a quantity not used to build them. Fourteen pre-registration documents (Supplementary S9, S18, S20, S21, S22, S24, S25, S28, S29, S30, S32, S33, S34, S35) each carry a reading table written before the run, and only their results register was completed afterwards; deviations are logged in the document they belong to. Five further documents sit beside that series and are not pre-registrations, which each of them says in its own header: a frozen decision record on the locus partition (S37); an estimand frozen except for one tolerance that remained an analysis choice (S38); a post-hoc exploratory control (S39); an external-validation protocol adopted but not registered, on three counts of its own — no qualifying cohort exists, no registry DOI was obtained, and no environment was locked beforehand (S40); and a transport test whose primary endpoint was fixed in advance but which we do not call pre-registered, because it carries no registry DOI (S41).

922 **Software**

923 R 4.4.1 with Seurat 5.5.1, Matrix 1.7.0, data.table 1.16.0, coloc 5.2.3, TwoSampleMR 0.7.5,
 924 susieR 0.14.2, arrow 25.0.0, hdf5r 1.3.12, TFBSTools 1.42.0, JASPAR2020 0.99.10,
 925 motifmatchr 1.26.0, chromVAR 1.26.0, BSgenome.Hsapiens.UCSC.hg38 1.4.5, survival 3.8.9
 926 and org.Hs.eg.db 3.19.1. Python 3.12.4 with numpy 2.0.0, pandas 2.2.2, scipy 1.18.0, pyarrow
 927 25.0.0 and matplotlib 3.11.1. External binaries: SMR v1.3.1 and PLINK v2.0.0-a.7.2. Ordinary
 928 least squares and Benjamini–Hochberg FDR were implemented directly in numpy rather than
 929 through statsmodels, whose installed version is incompatible with the installed scipy.

930 **Data and code availability**

931 All datasets are public and identified by accession above. Analysis code, intermediate result
 932 tables and the fourteen pre-registration documents will be deposited at [\(repository DOI\)](#),
 933 comprising every numbered analysis and figure script together with the tables needed to
 934 reproduce each figure and every number reported in the text. CQTNA, a runnable
 935 implementation of the diagnostics in the Discussion, is included in the deposit, as is a container
 936 definition pinning R, Python and the two external binaries to the versions used here. The
 937 analyses were run on Windows and the container is Linux, so it fixes the software versions and
 938 not the machine; the deposit states which numbers were checked across the two and which were
 939 not.

940 **Declarations**

941 **Ethics approval and consent to participate**

942 This study analysed only publicly available data: published or consortium-released genome-wide
 943 association summary statistics, published cis-eQTL summary statistics, and de-identified single-
 944 cell, spatial and chromatin datasets deposited in public repositories. No new human data were

945 collected and no individual-level participant data were accessed. Ethical approval and informed
946 consent were obtained by the original studies and are described in their primary publications and
947 repository records, which are cited by accession in the Methods.

948 **Consent for publication**

949 Not applicable.

950 **Availability of data and materials**

951 All datasets are public and identified by accession in the Methods. Analysis code, intermediate
952 result tables, the pre-registration documents and the container definition are in the deposit
953 described under Data and code availability.

954 **Competing interests**

955 <competing interests statement>

956 **Funding**

957 <funding statement, including grant numbers and the role of each funder in design, analysis,
958 interpretation and writing>

959 **Authors' contributions**

960 <author contributions, by initials>

961 **Acknowledgements**

962 <acknowledgements>

963 **Supplementary information**

964 S9–S51, including the fourteen pre-registration documents with their reading tables and results
965 registers; **S41**, the transport grid scoring six diseases against their own reference lists; **S42**, the

estimator identity, its numerical verification and the decomposition of the attribution by outcome significance; **S43**, the power-matched cross-disease comparison and the chance-corrected attribution statistic; **S44**, the seven-way method benchmark, run as a boundary check on the identity rather than as seven independent tests; **S45**, the fine-mapped decomposition and two failure modes of proxy-LD fine-mapping; **S46**, the diagnostic's own type I error, power and void rate; **S47**, the worked example in full; **S48**, the crossed grid cell by cell with its registration history; **S49**, the eight checks in full; **S50**, which inference unit the significance belongs to; and **S51**, how far the identity reaches into the audited literature, with the bounds that measurement admits. Earlier items include the literature-audit corpus with every PMID, both scoring passes, the two-coder subsample and the coding rules (S23); the multiple-testing unit sensitivity analysis, in which the significant list is recomputed over four units by minimum-p and by Simes (S26); the self-administered attribution check (S27); the complete record of target-substantiation attempts with the selection denominators and every stopping-rule instance, and the technical account of the two processing errors (S12); the replication-cohort search (S14); and the colocalisation window-and-prior sensitivity analysis (S15).

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1086 **Figures**

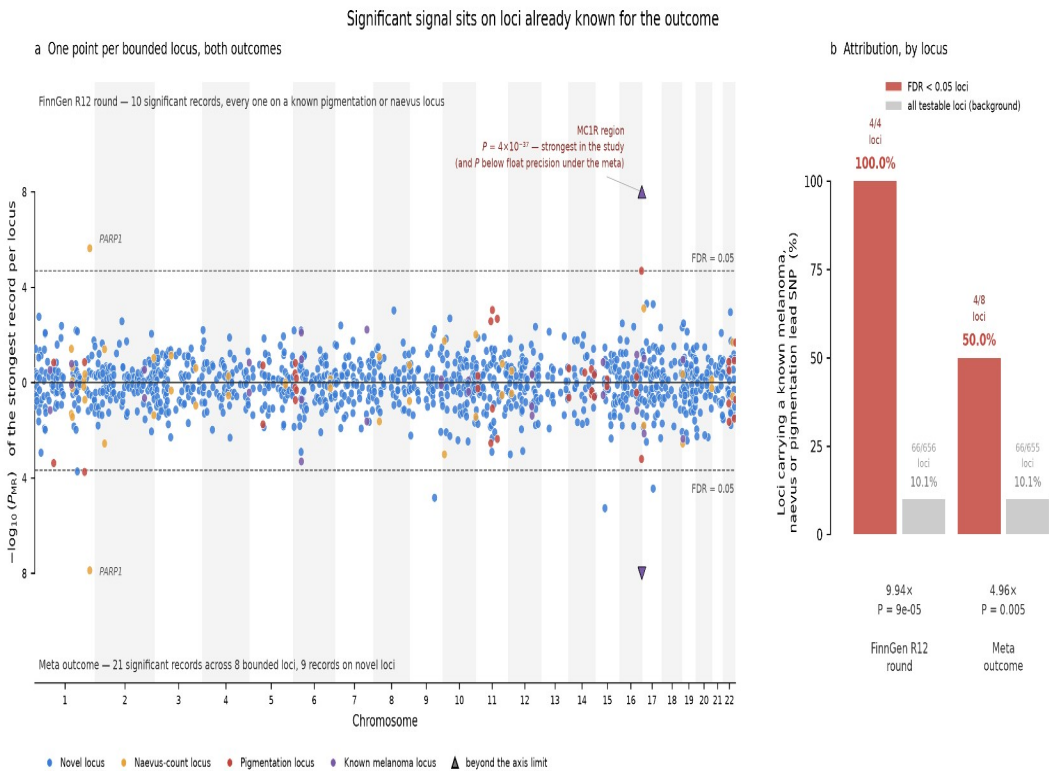


Fig. 1 | Locus attribution under both outcomes, by bounded locus. **a**, One point per bounded locus, both outcomes: a mirrored Manhattan plot in which each locus contributes its strongest record, FinnGen R12 above and the meta-analysis below, coloured by locus class. The MC1R region gives the strongest association in the study, $P = 4 \times 10^{-37}$, and PARP1 reproduces its published direction; these are the end-to-end positive controls. **b**, Attribution, by locus: for each outcome, the share of significant loci that are known loci against that outcome's own background share, 10.1% for the meta outcome. Loci are the non-recursive fixed-anchor partition frozen in Supplementary S37 — 1 Mb claimed from each anchor — not the single-linkage rule used in the source literature, which chains a dense resource into blocks of tens of megabases. Counting is by bounded locus throughout; the record-level version is Supplementary S10. [Fig1_locus_attribution.pdf/.png]

Changing one thing at a time: outcome power, disease, and exposure resource

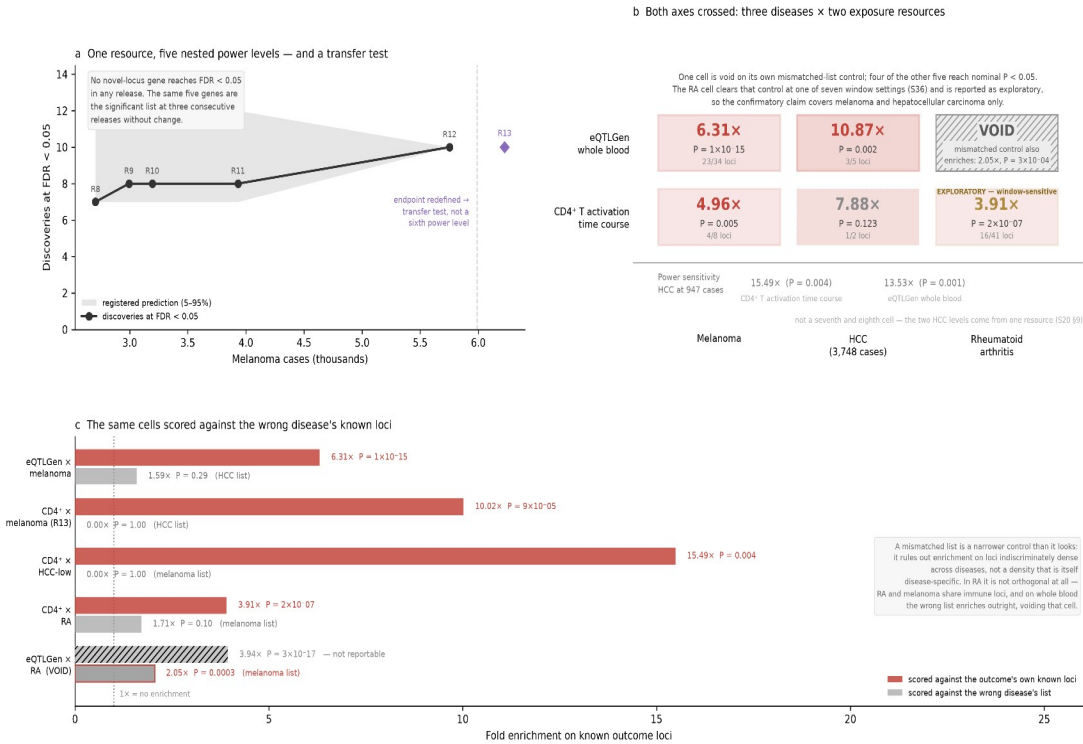


Fig. 2 | Generality: the attribution under one change at a time. **a**, One resource, five nested power levels, and a transfer test: FinnGen R8 to R12 carry an identical melanoma endpoint at rising case numbers, with everything else held fixed. R13 is separated by a rule and coloured apart because it is not a sixth power level — the endpoint definition and the control exclusions both change — and must not be read as a continuation of the curve. **b**, Both axes crossed: three diseases by two exposure resources. The second disease and the non-cancer outcome are two columns of this grid and the second exposure resource is its second row. The lower-powered hepatocellular level sits below a separating rule because it is a power sensitivity, not a seventh and eighth cell. **c**, The same cells scored against the wrong disease: each cell rescored with another disease's known-locus list. Without this panel, **b** can only show that significant loci fall on some loci, not that they fall on the loci of their own disease. All panels are computed on the frozen partition (Supplementary S37). [Fig2_generality.pdf / .png]

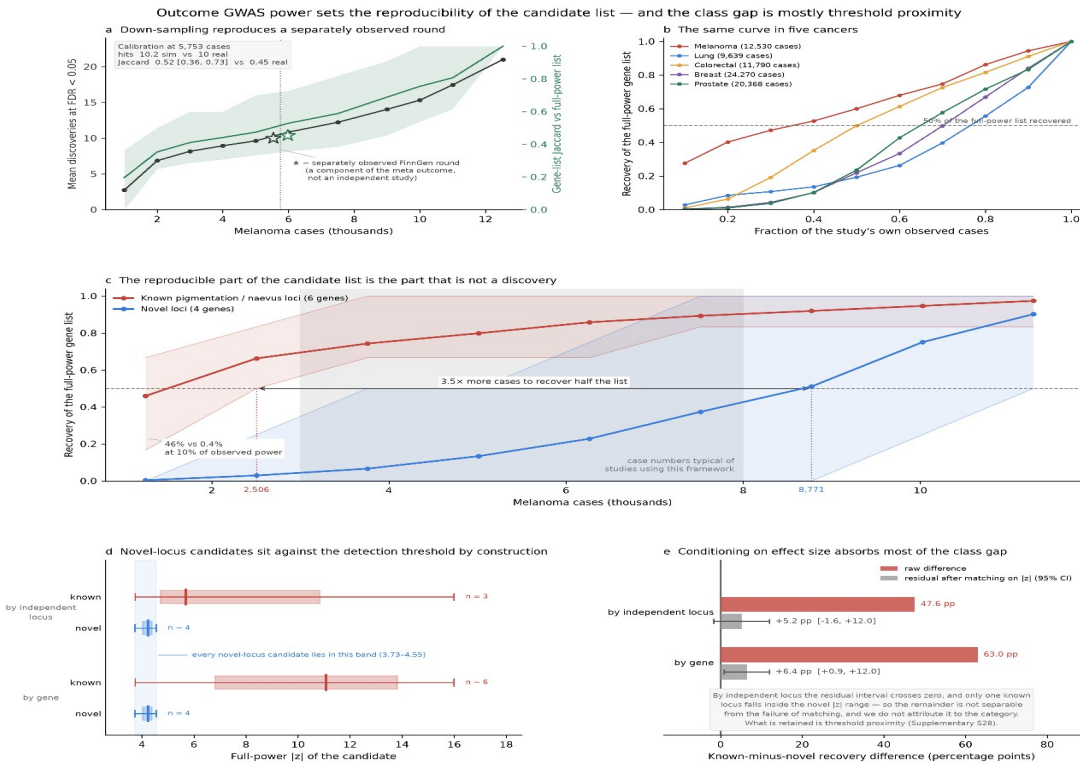


Fig. 3 | Power and list stability by locus class, and the same stratification against full-power $|z|$. **a**, Down-sampling reproduces a separately observed round, which is the positive control for the whole simulation. **b**, The same curve in five cancers. **c**, The reproducible part of the candidate list is the part that is not a discovery: recovery against power, split by locus class. **d**, Novel-locus candidates sit against the detection threshold by construction — their full-power $|z|$ all lie between 3.73 and 4.55, while known-locus candidates reach 15.99. **e**, Conditioning on effect size absorbs most of the class gap: raw gaps of 47.6 percentage points by bounded locus and 63.0 by gene fall to +5.2 and +6.4 after matching on $|z|$. The locus-level interval crosses zero, so **the residual is not claimed as an effect of the class label**; only one known locus lies inside the novel $|z|$ range, and that failure of common support is why. [Fig3_power_stability.pdf/.png]

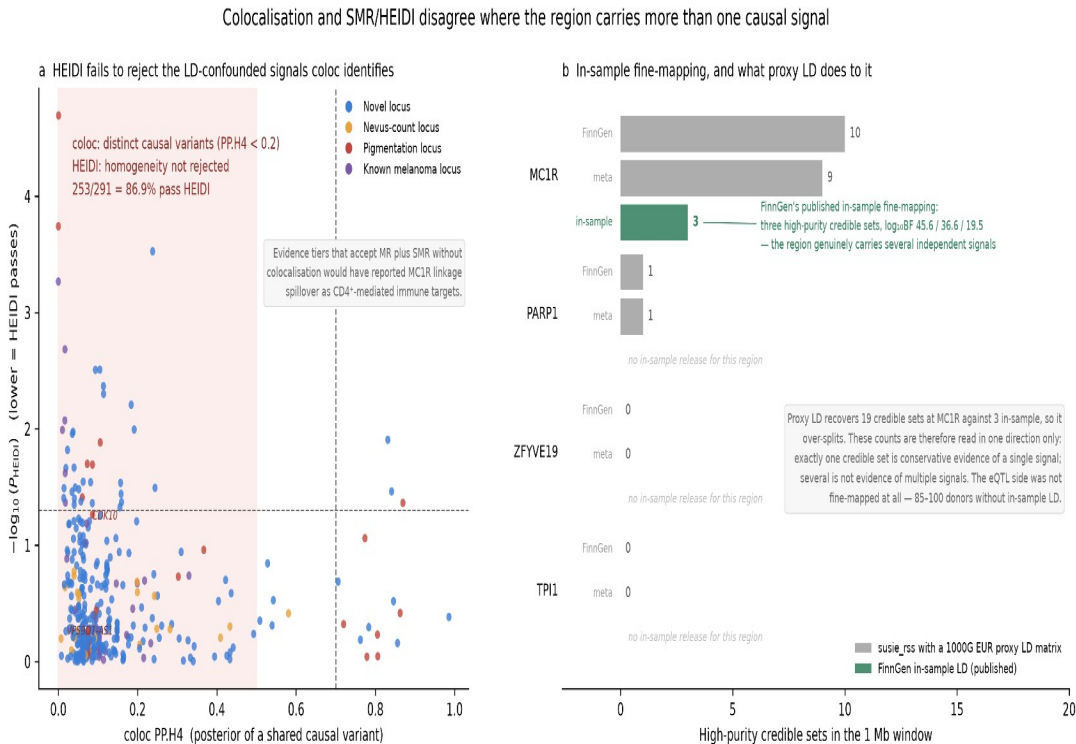


Fig. 4 | Colocalisation versus SMR/HEIDI, with in-sample fine-mapping. **a**, HEIDI fails to reject the LD-confounded signals colocalisation identifies: of the 291 records colocalisation assigns to distinct causal variants, HEIDI does not reject homogeneity for 253 (86.9%). **b**, In-sample fine-mapping, and what proxy LD does to it. FinnGen's published in-sample LD gives three high-purity credible sets at MC1R, so the region genuinely carries several independent signals — the configuration in which a neighbouring gene's eQTL is tagged by one of them without sharing it. The same region run on 1000 Genomes EUR proxy LD returns nineteen, which is over-splitting. **Proxy-LD credible-set counts are therefore one-directional: exactly one set is conservative evidence of a single signal, but several sets are not evidence of several.** In-sample counts exist for MC1R only and are not extrapolated to the other regions.

[Fig4_coloc_heidi.pdf / .png]

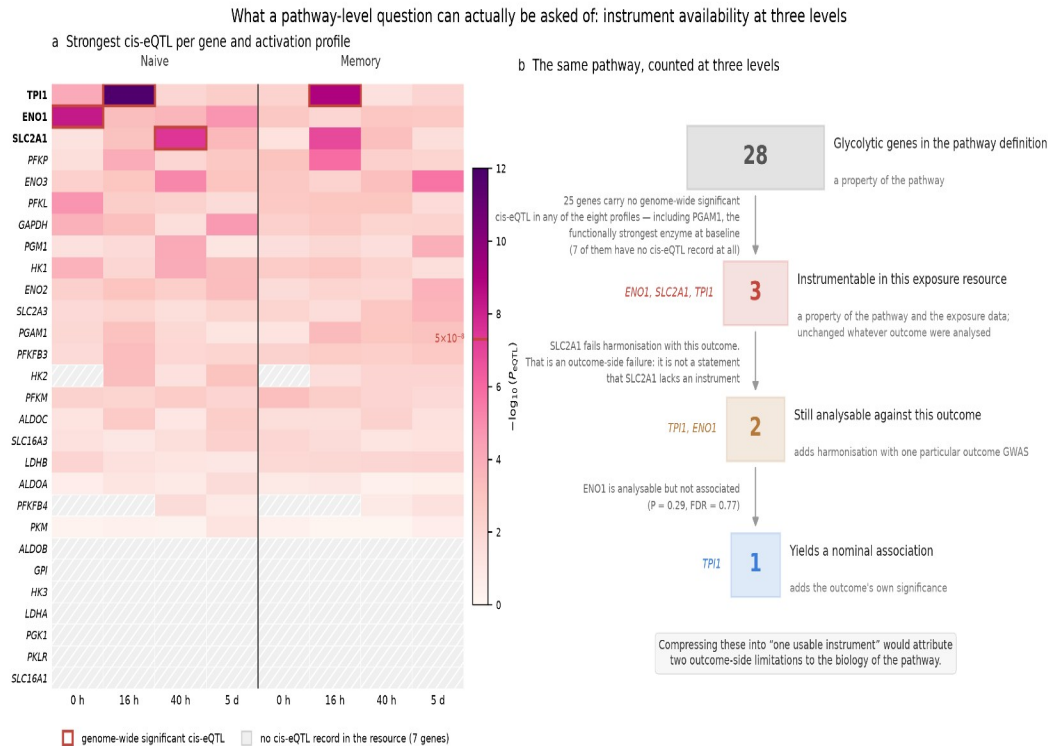
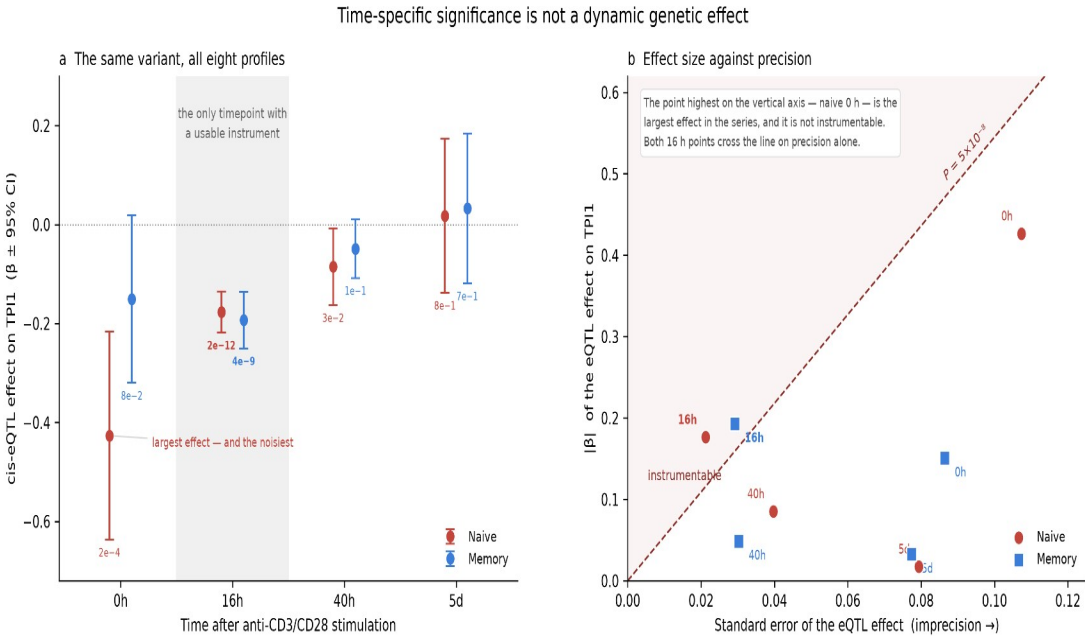


Fig. 5 | Instrument availability across the glycolytic pathway, at three levels. a, Strongest cis-eQTL per gene and activation profile: 28 glycolytic genes by 8 profiles, with the genome-wide threshold marked and genes absent from the resource shown separately. **b**, The same pathway counted at three levels, naming which gene drops out where and why. The three counts are not the same kind of statement and the panel keeps them apart: the first depends only on the pathway and the exposure data and is unchanged by the outcome; the second adds harmonisation against one specific outcome GWAS, so a gene lost there is not a gene without an instrument; only the third involves the outcome's own significance. [Fig5_instrument_ladder.pdf/.png]



What the window is not. In the published genotype $\times$ pseudotime interaction test [8], none of the three TPI1 lead variants shows an interaction — memory cells, linear $P = 0.974$ and quadratic 0.992; naive cells, 0.707 and 0.891 for the variant 1,063 bp from our instrument, 0.378 and 0.605 for the other — while TPI1 expression is among the most strongly pseudotime-dependent genes in that dataset (Moran's $I = 0.664$). What moves across the activation window is measurability, not regulation.

Fig. 6 | The activation window in which TPI1 is instrumentable: effect size against precision across the eight profiles. a, The same variant in all eight profiles, β with 95% confidence intervals. **b**, Effect size against precision. On this plane genome-wide significance is a ray through the origin, $|\beta| = 5.45 \times \text{SE}$. The resting profile sits highest on the effect-size axis and still falls below the ray: **the largest effect and the only usable instrument are not the same timepoint**. This is a statement about where the effect is measurable, not about where it is regulated, and it neither contradicts nor repeats the published genotype-by-pseudotime interaction test, which asks a different question and is negative at all three lead variants [4].

[Fig6_tpi1_window.pdf/.png]

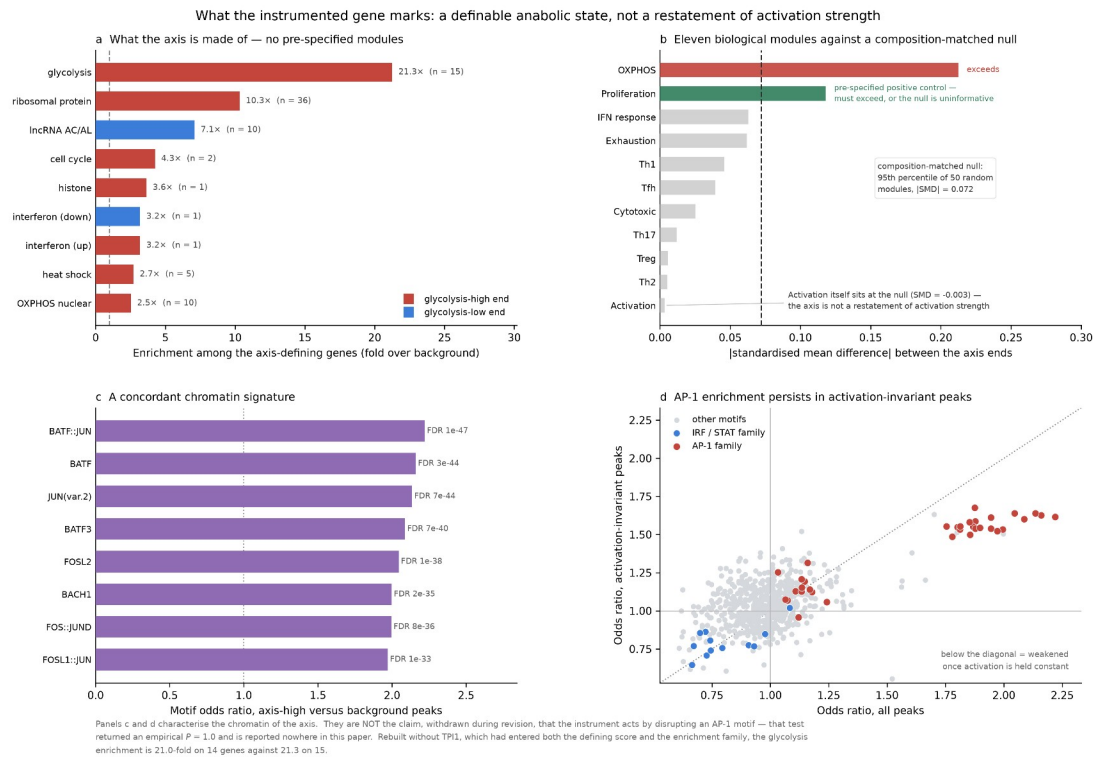


Fig. 7 | The CD4⁺ metabolic axis and its chromatin signature. **a**, What the axis is made of, with no pre-specified modules: family composition at the top of the residualised axis, glycolysis enriched 21.3-fold. **b**, Eleven biological modules against a composition-matched null, the threshold being the 95th percentile of |SMD| across fifty random modules. The activation module remained within the matched-null distribution. Two modules exceeded the threshold: OXPHOS, and the proliferation module, which was specified in advance as a sensitivity control (Methods). **c**, A concordant chromatin signature: motif enrichment in axis-high versus background peaks. **d**, AP-1 enrichment persists when peaks are restricted to those that do not change with activation. Panels **c** and **d** characterise the axis, and are not the withdrawn claim that the instrument acts by disrupting an AP-1 motif, which was tested separately and failed. [Fig7_axis_chromatin.pdf/.png]

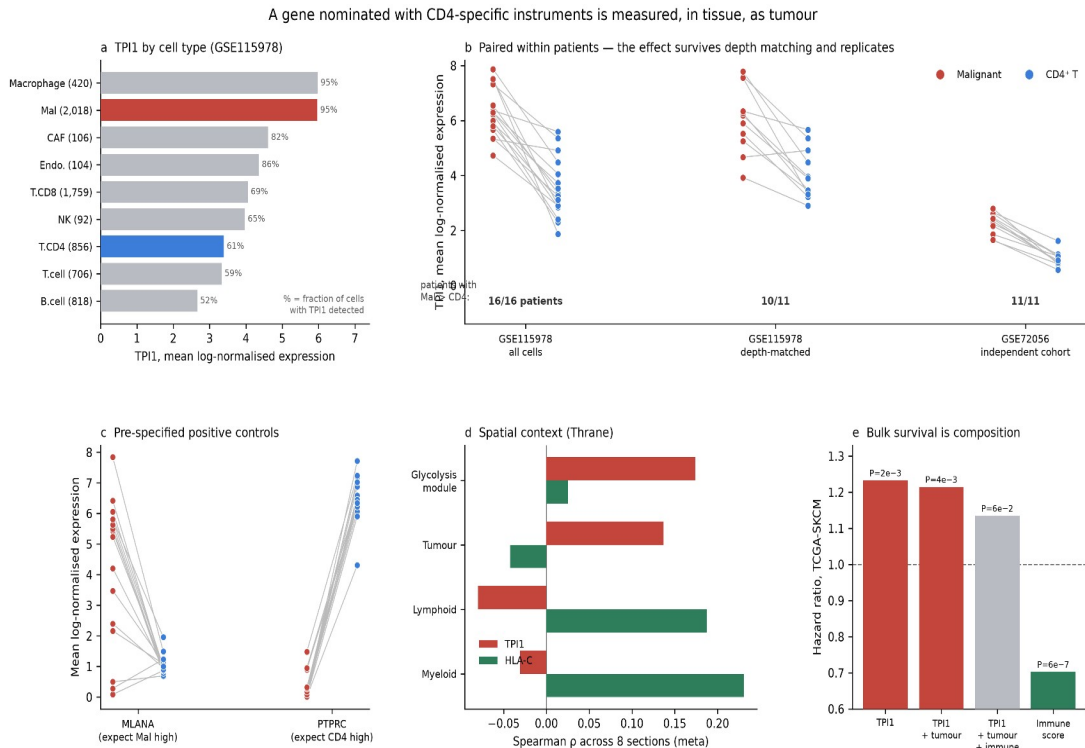


Fig. 8 | Compartment attribution. **a**, TPI1 by cell type, with detection rates. **b**, Paired within patients: malignant versus CD4⁺ cells, unmatched, depth-matched, and in an independent cohort side by side. This is the primary evidence in the figure. **c**, Pre-specified positive controls. **d**, Spatial context, as a cross-section meta-analysis. **e**, Bulk survival is composition, in TCGA-SKCM as distributed by Xena [14,15]. Spatial and bulk data are context here rather than evidence; the cell-level comparison in **b** is what the attribution rests on. [Fig8_compartment.pdf/.png]

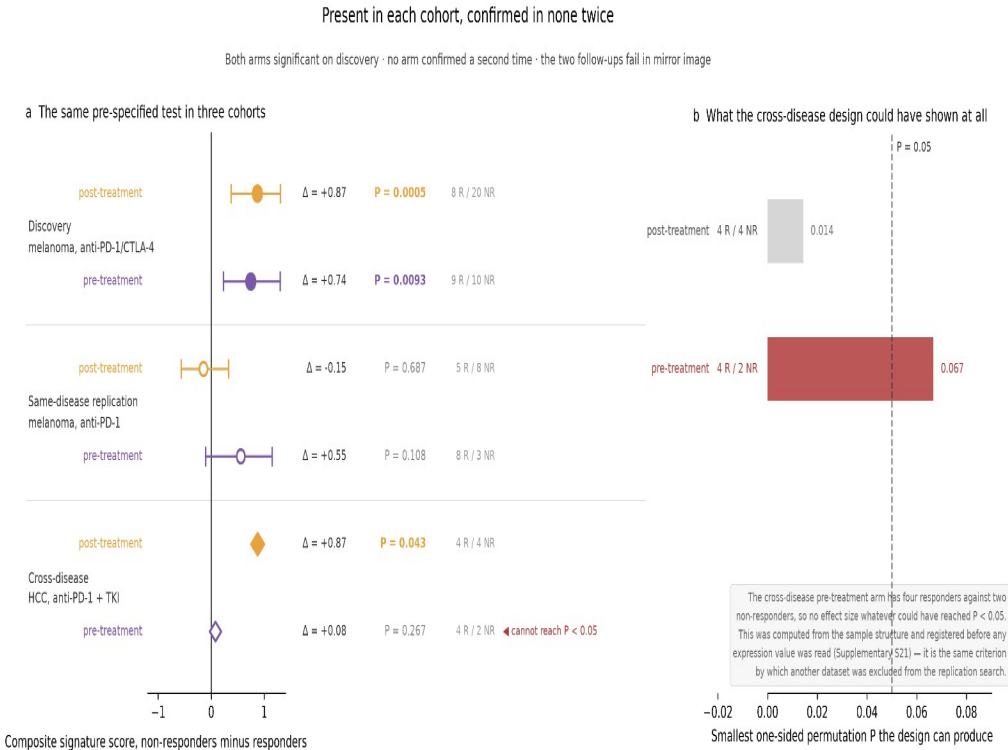


Fig. 9 | Patients across three cohorts. a, The same pre-specified test in three cohorts: effect size and one-sided permutation P for both arms. Both arms are significant in the discovery cohort and neither is confirmed twice, and the two failures are mirror images — the pre-treatment arm matches in magnitude in the same-disease cohort but is underpowered there and absent in the cross-disease one, while the post-treatment arm inverts in the same-disease cohort and returns at full magnitude in the cross-disease one. **b,** What the cross-disease design could have shown at all. Its pre-treatment arm compares four responders with two non-responders, so no effect size whatever can reach $P < 0.05$. That floor was computed and registered before any expression value was read (Supplementary S21), and a test that could not have been significant is not the same thing as a test that was run and was not significant. [Fig9_patients.pdf/.png]

Figure numbering follows this version's reading order and is independent of the numbering used in the full reference document. The self-administered attribution check is reported in the text and tabulated in Supplementary S27; it carries no main figure.